

Discretization

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Discrete
→

$$\begin{aligned}x_{k+1} &= A_d x_k + B_d u_k \\ y_k &= C_d x_k + D_d u_k\end{aligned}$$

Simple scalar system: $\dot{x} = ax \Rightarrow x(t) = e^{at} x(0)$

After timestep Δt : $x(t + \Delta t) = e^{a\Delta t} x(t)$

$$\dot{x} = Ax \Rightarrow x = e^{At} x(0) \Rightarrow x(t + \Delta t) = e^{A\Delta t} x(t)$$

$$\text{or: } x = e^{A\Delta t} x(t), \quad A_d = e^{A\Delta t} \Rightarrow \dot{x} = A_d x$$

Euler integration: $x_{k+1} = x_k + \dot{x}_k \Delta t$, $\dot{x}_k = Ax_k$

$$x_{k+1} = x_k + A x_k \Delta t \Rightarrow x_{k+1} = (I + A\Delta t) x_k \Rightarrow A_d \approx I + A\Delta t$$

↑
start of Taylor series for $e^{A\Delta t}$

To get B_d , solve $\dot{x} = Ax + Bu$

$$x(t + \Delta t) = e^{A\Delta t} x(t) + \int_t^{t+\Delta t} e^{A(t+\Delta t-\tau)} B u(\tau) d\tau$$

$$\text{If } u(\tau) = u_k \Rightarrow x_{k+1} = e^{A\Delta t} x_k + \int_0^{\Delta t} e^{A(\Delta t-\tau)} B u_k d\tau$$

$$B_d = \int_0^{\Delta t} e^{A\tau} B d\tau$$

Kalman Filter:

Prediction from model:

$$\hat{\mathbf{x}}_{k+1}^- = A \hat{\mathbf{x}}_k + B u_k$$

Sensor gives:

$$y_{k+1}$$

Can predict what we should have received from predicted state:

$$\hat{y}_{k+1} = C \hat{\mathbf{x}}_{k+1}^-$$

$$\text{Compare: } r_{k+1} = y_{k+1} - \hat{y}_{k+1} = y_{k+1} - C \hat{\mathbf{x}}_{k+1}^-$$

r is residual

Discrepancy between model and prediction is used to correct estimate:

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1}^- + K_k r_{k+1}, \text{ where } K_k \text{ is Kalman gain}$$

$$K_k = P_k^- C^T (C P_k^- C^T + R_k)^{-1} \quad \text{often referred to as } S$$

where P_k^- is uncertainty in predicted state

R_k is measurement/sensor uncertainty

Q_k is process/model uncertainty.

$$P_k^- = A P_{k-1} A^T + Q_k$$

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}^- + K_k r_k, \quad P_k = (I - K_k C) P_k^-$$